4090/s100/轮臂数据网络传输测试

一、测试背景

当前找人测试主要分为两种场景：

1. 轮臂+4090场景；----->基于agilex网络通讯
2. 真机（s100）+4090场景；----->基于iflyrobot网络通讯

因测试过程中需基于网络传输大量传感器数据，故进行相关网络传输效率测试（主要基于RGBD数据传输）。

二、总结

对于1280x720x15hz需求，单帧大小为2.76MB，15hz所需带宽为41.4 MB/s，总带宽理论上需要41.4x8 MB/s

1. 对于轮臂+4090场景，当前若传输非压缩数据，使用agilex网络时会出现无法传输数据的情况；
2. 对于s100+4090场景，当前在2.1与2.2场景（s100使用WiFi模块进行数据传输）中双相机同时使用，但读取帧率小于5hz，结合2.3与2.4场景s100使用有线数据传输效果，判断可能是WiFi模块素质导致传输效果不佳，需硬件同事确认；
3. 使用压缩数据传输效果及算法效果验证中，但D457当前驱动并不能给出压缩数据话题，压缩数据仅可用于轮臂效果验证；

代码块

```
/realsense/camB/color/image_raw
/realsense/camB/color/image_raw/compressed
/realsense/camB/depth/image_rect_raw
/realsense/camB/depth/image_rect_raw/compressed
```

三、测试内容

2.1 S100（WiFi）+ 4090（WiFi）+ iflyrobot

在不修改s100 UDP缓冲区大小（net.core.rmem_max = 212992）情况下，仅可在订阅两个RGBD相机（4*D457相机ROS2驱动均启动）情况下可传输数据，且数据存在降频，在开始订阅第三个相机时所有相机数据传输中断；数据传输帧率如下图所示：

```
average rate: 4.246
        min: 0.059s max: 60.939s std dev: 1.48524s window: 1677
average rate: 4.246
        min: 0.059s max: 60.939s std dev: 1.48303s window: 1682
average rate: 4.246
        min: 0.059s max: 60.939s std dev: 1.48083s window: 1687
average rate: 4.244
        min: 0.059s max: 60.939s std dev: 1.47908s window: 1691
average rate: 4.240
        min: 0.059s max: 60.939s std dev: 1.47779s window: 1694
average rate: 4.239
        min: 0.059s max: 60.939s std dev: 1.47605s window: 1698
average rate: 4.235
        min: 0.059s max: 60.939s std dev: 1.47477s window: 1701
average rate: 4.233
        min: 0.059s max: 60.939s std dev: 1.47304s window: 1705
average rate: 4.233
        min: 0.059s max: 60.939s std dev: 1.47131s window: 1709
average rate: 4.231
        min: 0.059s max: 60.939s std dev: 1.46917s window: 1714
average rate: 4.228
        min: 0.059s max: 60.939s std dev: 1.46790s window: 1717
average rate: 4.226
        min: 0.059s max: 60.939s std dev: 1.46619s window: 1721
```

```
average rate: 3.005
        min: 0.159s max: 61.023s std dev: 2.02138s window: 903
average rate: 3.008
        min: 0.159s max: 61.023s std dev: 2.01692s window: 907
average rate: 3.012
        min: 0.159s max: 61.023s std dev: 2.01250s window: 911
average rate: 3.016
        min: 0.159s max: 61.023s std dev: 2.00701s window: 916
average rate: 3.018
        min: 0.159s max: 61.023s std dev: 2.00264s window: 920
average rate: 3.021
        min: 0.159s max: 61.023s std dev: 1.99831s window: 924
average rate: 3.024
        min: 0.159s max: 61.023s std dev: 1.99401s window: 928
average rate: 3.027
        min: 0.159s max: 61.023s std dev: 1.98973s window: 932
average rate: 3.031
        min: 0.159s max: 61.023s std dev: 1.98442s window: 937
average rate: 3.036
        min: 0.159s max: 61.023s std dev: 1.97916s window: 942
```

修改UDP缓存区大小（256mb）后，反而出现无法接收相机数据，结合分析原因如下，后续测试中均不再对此进行调整：

代码块

```
1. 开发板快速发送 → UDP缓存积压更多数据（数百MB）
2. WiFi延迟突增(265ms) → 大量包堆积在缓存
3. FastDDS超时判定 → 触发重传
4. 重传包和原始包都在巨大缓存中排队
5. 队首阻塞 → 新数据被旧数据阻塞
6. 结果：延迟累积，最终帧率下降
```

2.2 S100（WiFi）+ 4090（网线）+ iflyrobot

在不修改s100 UDP缓冲区大小（net.core.rmem_max = 212992）情况下，仅可在订阅两个RGBD相机（4*D457相机ROS2驱动均启动）情况下可传输数据，且数据存在降频，在开始订阅第三个相机时所有相机数据传输中断；数据传输帧率如下图：

```
average rate: 3.256
        min: 0.091s max: 61.059s std dev: 2.26077s window: 1241
average rate: 3.258
        min: 0.091s max: 61.059s std dev: 2.25713s window: 1245
average rate: 3.259
        min: 0.091s max: 61.059s std dev: 2.25352s window: 1249
average rate: 3.261
        min: 0.091s max: 61.059s std dev: 2.24992s window: 1253
average rate: 3.263
        min: 0.091s max: 61.059s std dev: 2.24634s window: 1257
average rate: 3.258
        min: 0.091s max: 61.059s std dev: 2.24461s window: 1259
average rate: 3.252
        min: 0.091s max: 61.059s std dev: 2.24382s window: 1260
```

```
average rate: 1.273
        min: 0.201s max: 60.980s std dev: 4.23574s window: 347
average rate: 1.282
        min: 0.201s max: 60.980s std dev: 4.21191s window: 351
average rate: 1.287
        min: 0.201s max: 60.980s std dev: 4.19421s window: 354
average rate: 1.297
        min: 0.201s max: 60.980s std dev: 4.17109s window: 358
average rate: 1.303
        min: 0.201s max: 60.980s std dev: 4.15391s window: 361
average rate: 1.313
        min: 0.201s max: 60.980s std dev: 4.13142s window: 365
```

2.3 S100（网线）+ 4090（网线）+ iflyrobot

在不修改s100 UDP缓冲区大小（net.core.rmem_max = 212992）情况下，可订阅到四个相机传输的数据，但帧率与预期存在差距较大（相机数据频率均在5Hz左右）；数据传输帧率如下图：

```
average rate: 5.747
        min: 0.132s max: 0.198s std dev: 0.01060s window: 134
average rate: 5.732
        min: 0.132s max: 0.198s std dev: 0.01040s window: 140
average rate: 5.719
        min: 0.132s max: 0.198s std dev: 0.01012s window: 146
average rate: 5.706
        min: 0.132s max: 0.198s std dev: 0.01709s window: 152
average rate: 5.695
        min: 0.132s max: 0.198s std dev: 0.01763s window: 158
average rate: 5.684
        min: 0.132s max: 0.198s std dev: 0.01739s window: 164
average rate: 5.675
        min: 0.132s max: 0.198s std dev: 0.01714s window: 170
average rate: 5.665
        min: 0.132s max: 0.198s std dev: 0.01692s window: 176
```

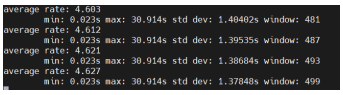
```
average rate: 5.549
        min: 0.132s max: 0.207s std dev: 0.01316s window: 227
average rate: 5.547
        min: 0.132s max: 0.207s std dev: 0.01301s window: 233
average rate: 5.543
        min: 0.132s max: 0.207s std dev: 0.01287s window: 239
average rate: 5.539
        min: 0.132s max: 0.207s std dev: 0.01273s window: 245
average rate: 5.536
        min: 0.132s max: 0.207s std dev: 0.01259s window: 251
average rate: 5.534
        min: 0.132s max: 0.207s std dev: 0.01246s window: 257
```

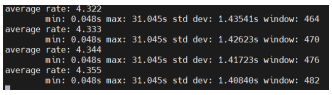
```
average rate: 5.464
        min: 0.133s max: 0.206s std dev: 0.00805s window: 249
average rate: 5.463
        min: 0.133s max: 0.206s std dev: 0.00796s window: 255
average rate: 5.462
        min: 0.133s max: 0.206s std dev: 0.00787s window: 261
average rate: 5.461
        min: 0.133s max: 0.206s std dev: 0.00779s window: 267
average rate: 5.460
        min: 0.133s max: 0.206s std dev: 0.00771s window: 273
average rate: 5.460
        min: 0.133s max: 0.206s std dev: 0.00763s window: 279
```

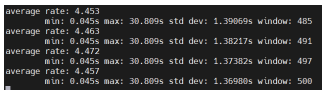
```
average rate: 5.398
        min: 0.178s max: 0.370s std dev: 0.01193s window: 246
average rate: 5.398
        min: 0.178s max: 0.370s std dev: 0.01178s window: 252
average rate: 5.399
        min: 0.178s max: 0.370s std dev: 0.01165s window: 258
average rate: 5.399
        min: 0.178s max: 0.370s std dev: 0.01151s window: 264
average rate: 5.400
        min: 0.178s max: 0.370s std dev: 0.01139s window: 270
```

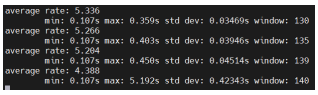
2.4 S100（网线）+ 4090（WiFi）+iflyrobot

在不修改s100 UDP缓冲区大小（net.core.rmem_max = 212992）情况下，可订阅到四个相机传输的数据，但帧率与预期存在差距较大（相机数据频率均在4Hz左右），且在重新订阅时会出现一段时间所有订阅器都数据中断的情况，运行过程中也会出现不定时的数据中断；数据传输帧率如下图：









2.5 轮臂（WiFi）+4090（WiFi）+ agilex

在轮臂启动单个D435i后，4090无法订阅到相机数据（同时使用笔记本连接agilex，同样订阅不到话题内容）

轮臂本地订阅数据在12.22下午3.30-4.40左右出现异常（数据帧率极低0.5-1hz），但在下午五点左右恢复正常，原因未知，过程中并未对网络或通讯做操作。